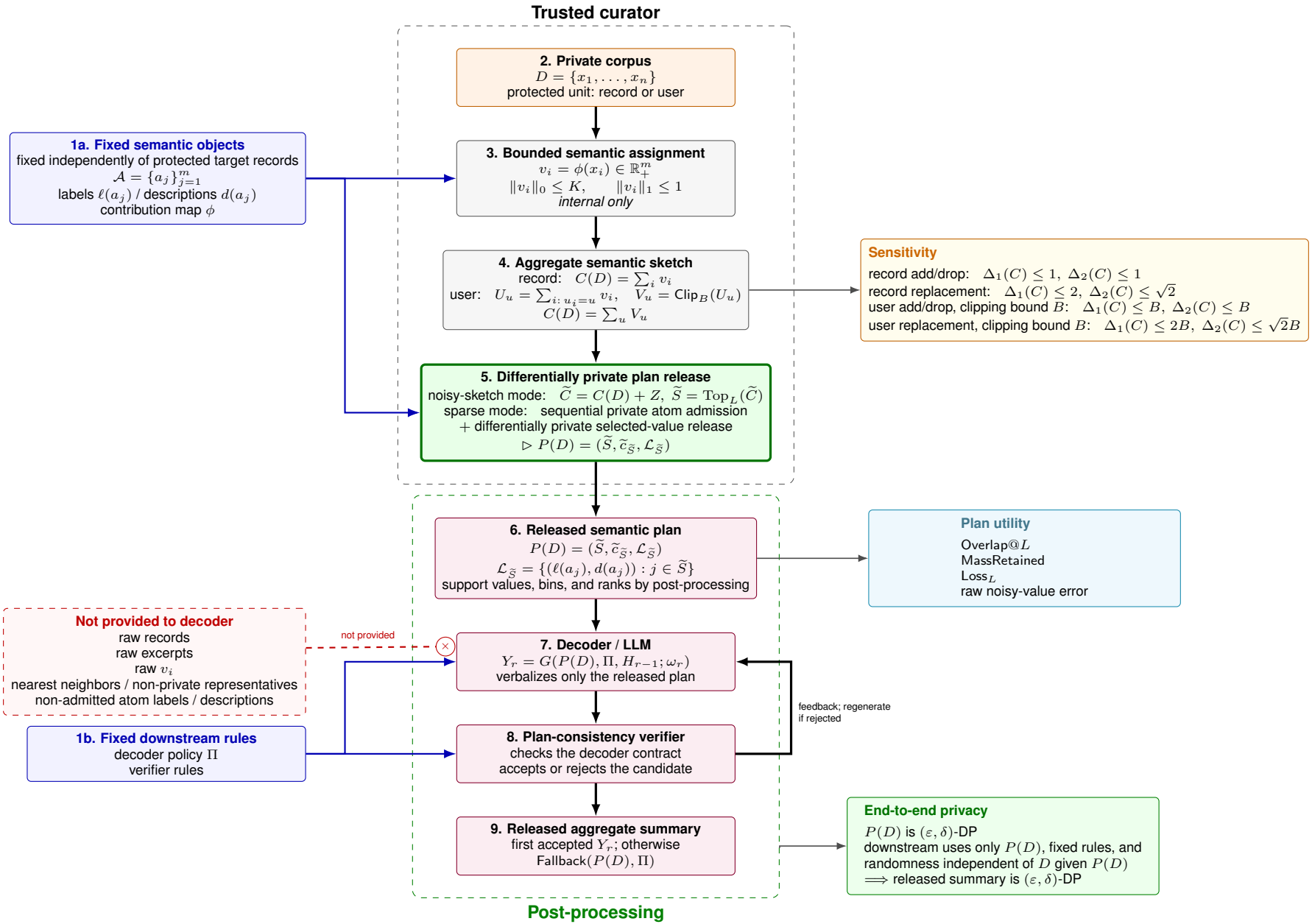

Trusted curator
2. Private corpus
$D = \{x_1, \ldots, x_n\}$
protected unit: record or user
1a. Fixed semantic objects
fixed independently of protected target records
$\mathcal{A} = \{a_j\}_{j=1}^m$
labels $\ell(a_j)$ / descriptions $d(a_j)$
contribution map ϕ
3. Bounded semantic assignment
$v_i = \phi(x_i) \in \mathbb{R}_+^m$
$\|v_i\|_0 \le K, \quad \|v_i\|_1 \le 1$
internal only
4. Aggregate semantic sketch
record: $C(D) = \sum_i v_i$
user: $U_u = \sum_{i:\, u_i = u} v_i, \quad V_u = \mathrm{Clip}_B(U_u)$
$C(D) = \sum_u V_u$
Sensitivity
record add/drop: $\Delta_1(C) \le 1,\ \Delta_2(C) \le 1$
record replacement: $\Delta_1(C) \le 2,\ \Delta_2(C) \le \sqrt{2}$
user add/drop, clipping bound B: $\Delta_1(C) \le B,\ \Delta_2(C) \le B$
user replacement, clipping bound B: $\Delta_1(C) \le 2B,\ \Delta_2(C) \le \sqrt{2}B$
5. Differentially private plan release
noisy-sketch mode: $\widetilde{C} = C(D) + Z,\ \widetilde{S} = \mathrm{Top}_L(\widetilde{C})$
sparse mode: sequential private atom admission
+ differentially private selected-value release
$\triangleright\ P(D) = (\widetilde{S}, \widetilde{c}_{\widetilde{S}}, \mathcal{L}_{\widetilde{S}})$
6. Released semantic plan
$P(D) = (\widetilde{S}, \widetilde{c}_{\widetilde{S}}, \mathcal{L}_{\widetilde{S}})$
$\mathcal{L}_{\widetilde{S}} = \{(\ell(a_j), d(a_j)) : j \in \widetilde{S}\}$
support values, bins, and ranks by post-processing
Plan utility
Overlap@L
MassRetained
Loss$_L$
raw noisy-value error
Not provided to decoder
raw records
raw excerpts
raw v_i
nearest neighbors / non-private representatives
non-admitted atom labels / descriptions
not provided
7. Decoder / LLM
$Y_r = G(P(D), \Pi, H_{r-1}; \omega_r)$
verbalizes only the released plan
feedback; regenerate if rejected
1b. Fixed downstream rules
decoder policy Π
verifier rules
8. Plan-consistency verifier
checks the decoder contract
accepts or rejects the candidate
9. Released aggregate summary
first accepted Y_r; otherwise
Fallback$(P(D), \Pi)$
End-to-end privacy
$P(D)$ is (ε, δ)-DP
downstream uses only $P(D)$, fixed rules, and
randomness independent of D given $P(D)$
$\Longrightarrow$ released summary is (ε, δ)-DP
Post-processing